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A Systematic Review on Urban Road Traffic Congestion

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Abstract
The city's infrastructure is considered the backbone of any country's development process and there are numerous factors that contribute to its growth. Among these factors, proper management traffic management is crucial. The increasing traffic density poses challenges to the current infrastructure, especially in developing countries, leading to issues such as congestion and security. Technological advancements have introduced intelligent transportation systems that offer innovative mobility solutions and promote sustainability. To provide better solutions, a systematic review was conducted following the PRISMA rules. Three electronic databases, namely IEEE Xplore, Science Direct, and Wiley, were searched using specific keywords. Research articles were identified, accessed, and included in the review based on the PRISMA rules. This systematic review explores various approaches used for predicting, detecting, and analyzing congestion levels on urban roads. These approaches are categorized based on their datasets, results, and comparison with other available algorithms. Additionally, the discussions expand on the advantages and limitations of different categorical approaches.

Keywords Intelligent transportation systems · Traffic congestion · Urban roads · PRISMA rules · Systematic review

1 Introduction

Congestion has always been a significant concern in traffic control mechanisms caused by an unbalanced demand–supply situation that directly affects the problem of a large amount of fuel consumption and greenhouse emission. Usually, in densely populated conditions other than traffic jams, unhealthy air quality, vibrations, noises, and accidents are significant issues for quality control authorities. T. Thomas explained these challenges that are threatening expeditions because of speedy increases in traffic [1]. In different countries, depilating infrastructure outright and more transport on the road presents a unique and highly perplexing jam. This presents a unique challenge in traffic management.

An increment in traffic demand directly affects the perpetual traffic on roads in densely populated areas where traffic roads are the primary source of conveyance for inter-city, and intra-city are directly hit by congestion. Many factors that may lead to this congestion problem are accidents, bad weather conditions, the slow movement of heavy vehicles,

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sudden failure of vehicles, negligence of not following traffic rules, sudden jams at road intersections and physical conditions.

The manner of commuters is also a genuine issue. One of the reasons might be that people get driver's licenses without taking practical driving tests. Due to a lack of appropriate learning of driving rules and regulations, people do not seem well known about these. Commuters do not care about such rules and intend to reach their destinations without following the traffic rules. This behaviour causes more delays and, in turn, produces traffic congestion. Even most people do not maintain the lanes and frequently attempt to move their transports from one lane to another.

The innovative concept of intelligent transportation under the domain of smart cities solves the congestion problems arising due to the lack of resources or lousy road infrastructure. Smart cities underlying many emerging techniques that are applied for controlling different factors smartly under diverse domains, e.g., energy management, medicine, surveillance, disaster management, security, home automation, and transportation, as shown in Fig. 1. These innovative technologies have mainly contributed to converting urban cities into smart cities. These intelligent systems provide users and city management to provide all facilities with limited resources to maintain their integrity in different aspects of operations such as traffic control, good quality of life, green emission, energy management, etc. Under these domains, smart transportation is vital to the success of the implementation of future endeavors in growing infrastructure.

According to the US Department of Transportation, "Intelligent Transportation Systems (ITS) apply a variety of technologies to monitor, evaluate, and manage transportation systems to enhance efficiency and safety" [2].

Smart city traffic transportation is revolutionizing by different approaches that how they respond the mobility and emergency response while mitigating congestion on city streets with communication technologies, sensors, intelligent systems, or high-speed networks [3]. Afrin [4] initiated that this technological growth can be badly hurt by uncertain congestion factors. Traffic congestion is a condition in conveying that is characterized by more

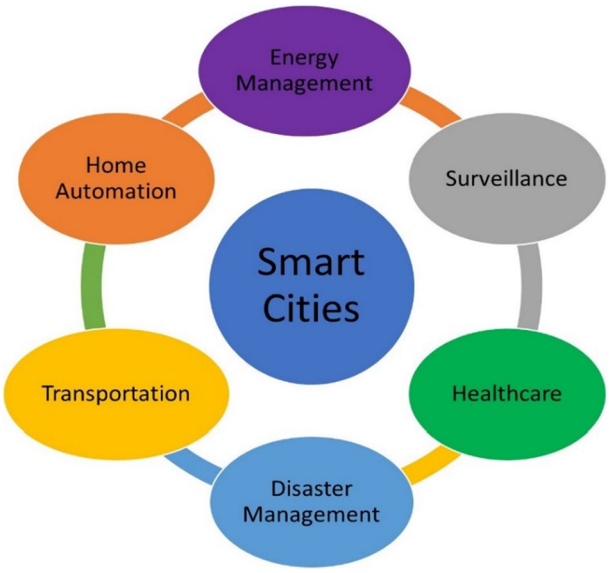

Fig. 1 Commonly implemented smart city domain

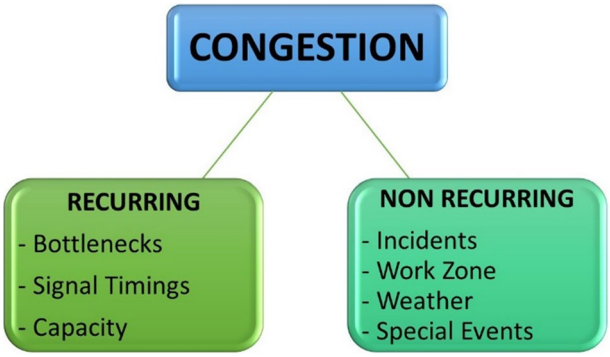

Fig. 2 Classification of congestion types [5]

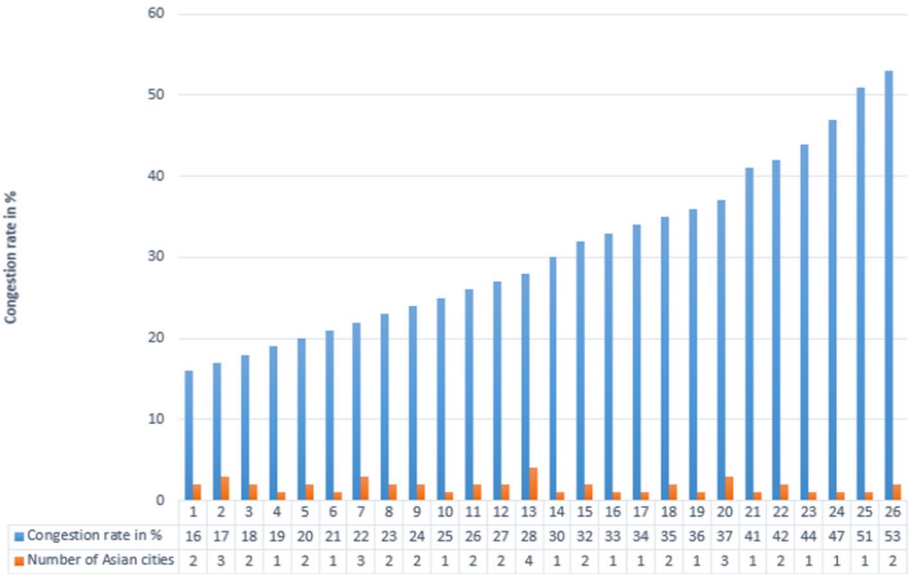

Fig. 3 Congelation rate in Asian Countries [6]

variable speeds, longer trip times, and incremented vehicular queueing. Advanced countries have already adopted solutions to overcome the problem of congestion.

Traffic congestion in particular is divided into two categories as recurring and nonrecurring congestion. Recurring congestion results from demand exceeding supply over a certain time or space. Non-recurring congestion customarily is caused by infrequent events, e.g., incidents, work zones, and special events as shown in Fig. 2, [5]. TOMTOM published a survey based on congestion levels for about 57 countries in 6 continents that collected data from different resources and divide them city-wise for different congestion factors which shows that Asian countries are mostly affected by this traffic congestion problem as illustrated in Fig. 3 [6].

A rising figure of the vehicle on roads is very concerning for any country's economy, infrastructure and environmental condition; more focused on healthy living conditions, an ordinary person can get badly affected. To address these issues efficiently, a comprehensive review is made of existing methods of congestion prediction levels on urban and highway

roads. This paper gives a clear and comprehensive review for researchers and also highlights the methodological approaches for congestion prediction. Following are the research objectives of this review paper:

1. To categorize the traffic congestion prediction techniques into five major classes as heuristic approaches, sensing base approaches, artificial intelligence approaches, wireless sensor base approaches, and scheduling approaches.
2. To present a summarized review of key research papers for the last eleven years from three different databases.
3. To analyze the current research work and extant the future direction for research with advanced methods to improve the congestion prediction accuracy.

The rest of the paper is organized as follows. In Sect. 2, methodology is discuss. Section 3 discusses the different approaches for prediction techniques, Sect. 4 deals with the discussions and Sect. 5 discusses the conclusions.

2 Methodology

In this section the methodology for research review have been describe. Section 2.1 describes the database collection, Sect. 2.2 contains the inclusion criteria of research papers and Sect. 2.3 discusses the exclusion criteria of research articles that have not been selected.

2.1 Database Collection

The systematic review is carried out according to the selected research papers and their analysis [7]. The literature review was performed and investigated for relevant knowledge. Three online databases of IEEE Xplore, Science Direct, and Wiley were searched by using the keyword of ***URBAN TRAFFIC CONGESTION*** for the period last eleven years from 2011 to 2021. The total research articles identified were 211,210 including journal articles and conference proceedings from three databases that includes 190,407 publications from IEEE, 2564 from Wiley, and 18,239 from Science Direct.

The selection criteria are journal publications; therefore 67,511 conference proceedings are not selected; and total of 143, 699 research papers are screened; out of which 420 research papers have been selected that are closely related to road congestion. These research papers were examined carefully and 70 research articles have been selected that only focusses urban road traffic congestion; the categorical distribution according to the number of research papers is shown in Fig. 4; and the selection procedure following the PRISMA rules is shown in Fig. 5.

2.2 Inclusion Criteria

Relevant research papers have been included through the following criteria:

- Research papers published between 2011 and 2021.
- Research paper published in English.

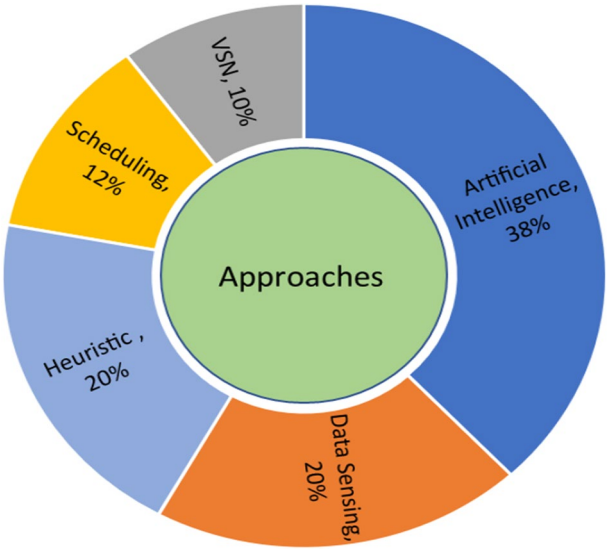

Fig. 4 Categorical distribution of approaches according to the number of research papers included in review

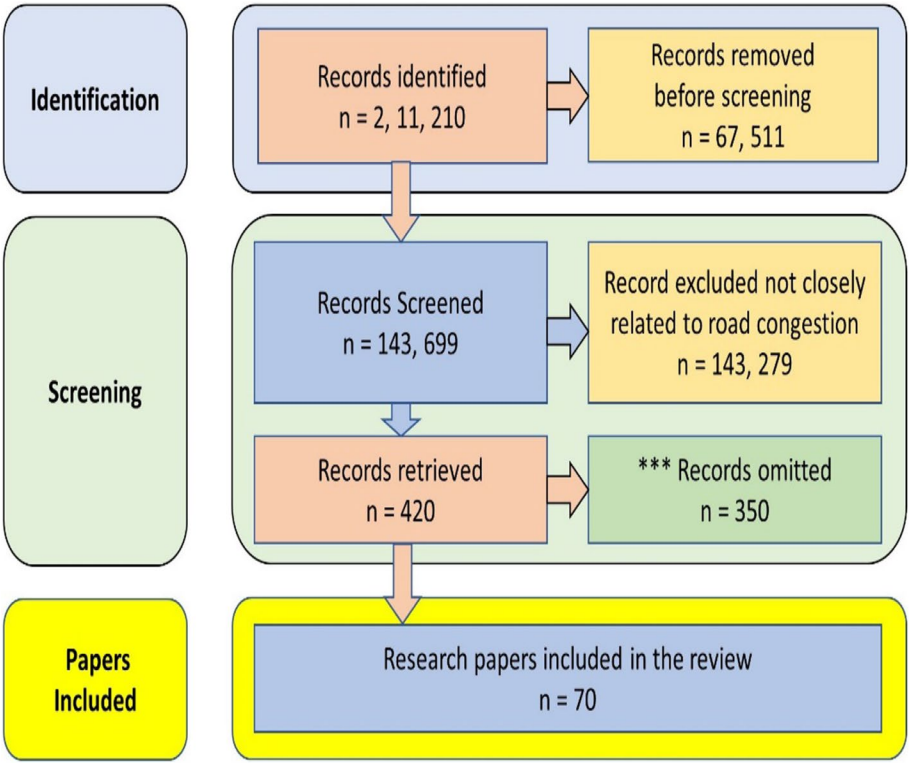

Fig. 5 Prisma Flow Diagram illustrating the procedure of extracting desired research paper

Table 1 Shows the summary of Heuristic approaches for the prediction of congestion levels for urban road traffic in the last 10 years

Reference	Algorithms/methods	Data	Results	Limitations
[6]	Autoaggressive model	Data collected from 20 signalized intersections in the city of Almelo, Netherlands	The 24 h prediction is more accurate than baseline predictions The error in short term predictions are negligible due to the extrapolation mechanism	The error in short term predictions are considered due to extrapolation mechanism
[10]	NRR[a]	Sub-set of TAPAS-Cologne 0.17.0 is used for evaluation	NRR[a] reduce the average trip time reliability up to 38.02% and improvement in reliability of travel time up to 65.42%	Average travel time is increased by 28.94% thus planning time index is raised by 80.99% yields unreliable results
[11]	SVRCACO[b] algorithm	Peak hours data is collected from Civil Motorway in Panchio City, Taiwan	SVR[c] models predict good efficiency over the [4]ARIMA model	Different parameters speed limitations, heavy vehicles, bottleneck and waiting time are not addressed
[12]	SWARIMA[d]	Intersection data for 12 h in the City of China for morning and evening peaks	SWARIMA[e] model results predict three predictive levels of warning time, congestion, and mitigation for an elastic interval of times	Cannot predict outside the allowable error range
[13]	HMM[f]	Road traffic state and internal road traffic state in the hospital area of Ningbo City, China	Prediction accuracy is 5.8% than the auto-regressive model	No validation of algorithm Limitation of when data is not affected by external circumstances Distance limited prediction
[14]	Moran's I autocorrelation method	Data gathered from more than 3000 taxis in Chengdu, China	Low percentage of 4.17% for false-positive values Good prediction in peak hours	Dissipation space due to overall convergence rate to congestion is weak
[15]	HMM[f]	IoT Traffic Sensors in Denmark	Prediction accuracy is 89%	-Not accurate results with less neighbouring roads
[16]	Nonparametric regression model, Fourier history estimated model, nonparametric autoregressive model, and nonparametric neighborhood regression model	Data is obtained from expressway in between Huayuan North Road and Xueyuanqiao Bridge, Beijing, China	MAPE[g], MAE[h], and MaxAE[i] reflects the good prediction for the developed model	Complex model Traffic volume for different roads are missing
[17]	MDF[j]	500 Taxi trip data in the megacity of China	Sharing decreases service times and maximize the coverage	Interaction and intersections problems are not addressed Missing detailed modelling for passenger modes of selection

Table 1 (continued)

Reference	Algorithms/methods	Data	Results	Limitations
[18]	EM[k]	Disaggregated data from LA, UK	Less complex model than existing EM[k] models No proper management to address the effect of CO_2 on congestion	All traffic variables are not fully utilized that can increase the congestion accuracy and emission prediction
[19]	Large neighborhood search	Data is collected in California, the USA from 426 links and 267 nodes with a cycle length of 100 s	Passenger hour travel time improve between 15 to 19%	Lack of connectivity constrains
[20]	SVD[l]	Three months data is collected in 446 zones in Shanghai, China	Congestion in peak hour returned to normal value	Travel mode transfer during and after COVID-19 is not considered
[21]	VT-Microscopic Modelling	Data was collected from subnetwork 1–5 in the city of Los Angeles, subnetwork 6 in Doha Qatar, and subnetwork 7 in Washington DC USA	Reduction of 15% in traffic demand Reduction of 60% in delay	Environmental parameters are not considered
[22]	Trans-logarithmic algorithm	Data collected from the Ministry of transport, Malaysia adjusted with annual growth	Result analysis for the developed model is best against Greenberg and Greenshields models	Non-statistical traffic characteristics are not considered

This table covers algorithms, data collection places/sources, and results in terms of accuracy and comparisons with different models where available

[a]Next Road rerouting

[b]Support vector regression model with continuous ant colony optimization

[c]Support Vector regression

[d]Sliding Window Autoregressive integrated moving average

[e]Autoregressive integrated moving average

[f]Hidden Markov Model

[g]Mean absolute percentage error

[h]Mean Absolute error

[i]Maximum Absolute error

[j]Microscopic fundamental diagram

[k]Emissions models

[l]Singular Value Decomposition

- Peer-reviewed publications preferably in a Journal.

2.3 Exclusion Criteria

Exclusion criteria have been formulated as follows:

- Studies that do not focus explicitly on Urban Road Congestion.
- Research papers published before 2010.
- Removal of duplicate records identified through endnote.
- Objectives was missing.
- Publication of studies in other languages other than English.

3 Categorical Approaches for Prediction of Urban Road Traffic Congestion

Urban traffic congestion has become a significant problem in the development process of any country and therefore needs to be addressed effectively. Different approaches have been adopted to measure traffic congestion by collecting relevant traffic information. The researchers have adopted different approaches based on data collected from any physical source or any dataset available from relevant traffic authorities. These proposed approaches are identified according to implemented methodology, data collection sources, and results. These adopted approaches are examined, identified, and classified. Among these approaches, Heuristic approaches are considered problem-solving approaches that are utilized in situations where there are constraints of time limits but require speedy processes for a quick decision.

In contrast, Artificial intelligence approaches have gained tremendous fame in the recent decade because their accuracy and improved performance are well-cooperated with systems for continuous predictions. Scheduling approaches are optimized approaches that are simple and adaptive to prevailing traffic conditions. The other two adopted approaches have different applications based on data collection sources; the data sensing approaches gathered traffic information from sensors available on roadsides, and the Vehicular Sensor Network used onboard sensors deployed on cars to collect traffic information and transmit it to central units for processing.

3.1 Heuristic Approaches

Urban traffic congestion was addressed by different heuristic approaches to address the problems by implementing different techniques as summarized in Table 1.

A prediction method was proposed to minimize the systematic variation in traffic states [1]. Wang et al. [8] proposed a system to assist drivers to avoid unexpected congestion by reducing the average trip time. To improve the efficiency of inter-urban traffic flow in peak hours and to avoid congestion an investigation was performed by support vector regression model [9]. Another smart approach was adopted that accepts the real-time traffic data stream and analyzes the elastic time interval for the prediction of congestion trends [10]. Sun et al. [11] effectively produced a congestion pattern prediction model that creates a correlation between the hidden and observation states for a busy traffic zone. Chen et al. [12] predicted the congestion states for Spatio-temporal correlation characteristics of traffic

by improving the mixed forest prediction method by adding the parameter of common congestion coefficient. Priambodo utilized Hidden Markov modeling to predict the impact of congestion on neighboring roads [13]. To avoid congestion using traffic states; a forecasting model was developed by adopting the combination of non-parametric regression models [14]. Caio et al. [15] studied the effect of growing fleet sizes of the transportation network and proposed a method that depends on the willingness to share and operational approaches to avoid congestion. Matt [16] proposed an approach of emission model to verify the inclusion of CO_2 in urban traffic congestion. Dimitrios et al. [17] provided the optimal solution that addresses the dynamic congestion problems by modeling framework through the implementation of a large neighborhood search method. A retrospective analysis was performed using a singular value decomposition approach by observing the change in speed rate and identify the variance for different traffic conditions during the COVID-19 pandemic [18]. Jian Du et al. [19] VT-Microscopic modeling was performed to study the effect of reduced emission, fuel consumption, and vehicle delays during the COVID-19 pandemic that significantly showed the decrease in traffic delay. Mustapha et al. [20] used the trans-logarithmic method to find the appropriate models for selected traffic congestion areas by using traffic states relationship.

3.2 Data Sensing Approaches

Researches have been further extended by gathering data from physical sources implemented on roads and highways as illustrated in Fig. 6. These data are analysed and implemented through different algorithms as summarized in Table 2. Chu et al. [21] presented a traffic flow model to avoid congestion by observing the evolution of traffic flow on the highway and found that the error chances for prediction are significantly decreased for different local detectors on-road. Nur et al. [22] proposed a solution for heavy traffic congestion by an improved automated traffic control system that can make an optimal decision for on different existing algorithms. He et al. [23] used a cell transmission model (CTM) for reproducing broken links in upstream and ramp flow in densely populated areas traffic. An empirical analysis was performed for predicting short term traffic flow method by observing delay and non-linear properties of urban traffic [24]. Jo et al. [25] developed an effective method based on real-time traffic information for the variable speed limit to enhance the effectiveness of traffic at multiple stations. Wang et al. [26] developed a visual

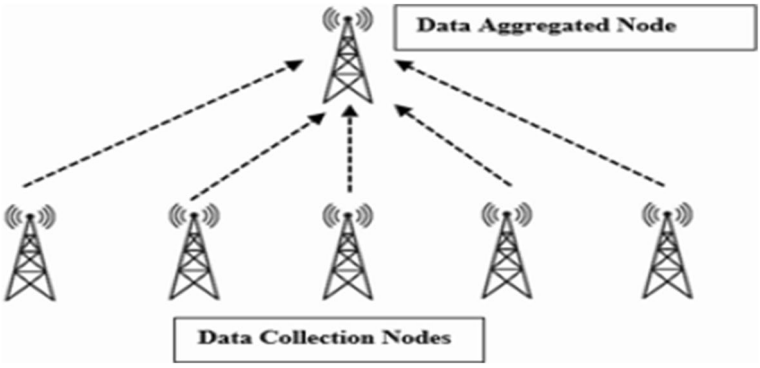

Fig. 6 Data sensing approach

Table 2 Summary of data sensing approaches for urban traffic road congestion for last ten years

Reference	Algorithms/methods	Sensors	Data	Results	Limitations
[23]	SPDE[a]	23 Inductive Loop Detectors	Data is collected from Northeastern Virginia and Washington DC	The proposed SPDE[a] model yields a good result with more predictive power for variance in traffic flow daily	Both SPDE model and simulation were calibrated with same traffic data Simulation with SPDE model have 9–15% error but can be high
[24]	ACRTCS[b] Algorithm	Remote traffic microwave sensors	Data Collection place not mentioned	The proposed ACRTCS[b] system reduces the average waiting time for vehicle	Limitation of data availability due to limited observation time interval of 900 s
[25]	CTM[c]	Barometer detectors	Collection of data from a single segment in between Xizhimen to Deshengmen, Beijing 2nd Ring road, China	Successful reproduction of induced and spontaneous traffic breakdown	Model didn't defined the transition state in between free flow and congested states
[26]	PSO[d] algorithm, Non-linear grey algorithm	Camera for video analysis	A single section of Youyi Avenue between Yuanlin Road and Jianshe one Road of Wuchang District of Wuhan	Measured values are closed to forecasted values for the Grey model Effectively address the delay and non-linear traffic characteristics	Uncertain conditions of traffic didn't cater
[27]	Variable speed limitation	Detectors not specified	Collection of data in 11 entrance/exit gates at the section between Co RD 42nd to 113th station in Minneapolis, Minnesota, USA	Speed variance is reduced to 14.2% to ensure good control of speed Average maximum speed is reduced to 20.1% in between different station	Irregular distances are not addresses in VSL algorithm
[28]	STM[e]	GPS Sensors	Trajectories data of 28,519 Taxis collected in Beijing, China	High-level pictures of congestion with propagation graphs and on-road segment level	No real time traffic predictions
[29]	ECTM[f] Dijkstra's algorithm	Detectors not specified	Data Collection place not mentioned	Predicted results show that 80% of drivers receive prior information of accidents that are on the same route	Dynamic prediction for different traffic density Accurate estimation of time for obtaining shortest path

Table 2 (continued)

Reference	Algorithms/methods	Sensors	Data	Results	Limitations
[30]	STCS[g]	GPS Sensor	Taxi trajectory data of over 12,000 taxis for 113,765 road segments are collected in Shanghai, China	Detection of congested segment regardless of time interval	High complex structure due to use of multiple kernels
[31]	ASM[h] Dividing Rectangles DIRECT method	Remote microwave sensors, license plate recognition cameras	Data collected from the Traffic Management platform of Hangzhou, China	Real-time prediction to improve the accuracy	No integration of traffic flow conservation law Didn't consider the change in volume of traffic during predictions
[32]	BFS[i] algorithm Markov Analysis	Inductive loop detectors	Data is collected in Taipei, Taiwan	Accurate prediction of bottlenecks due to congestion level and propagation in comparison with other methods	Fine grained cost of traffic congestion levels are not taken into account
[33]	Index calculation and clustering method	GPS sensors for FCD data	Rainfall and [22]FCD[j] Data is acquired from the Transport Commission of Shenzhen, South China	Speed reduced by 6.20% on weekdays that increase traffic congestion by 23.53% Speed reduced by 2.37% on weekends with an increase in traffic congestion by 20.65%	Traffic fluctuations and accidental data is not available
[34]	The dynamic user equilibrium method	Loop Detectors and detection cameras	Data is gathered from 30 segments in El Paso, the city in Texas, USA	Generation of resilience curve for the first time as it was just a concept Different Resilience curves indicate the priority links for immediate clearance of incident conditions	Traffic metric didn't correlate to form traffic index
[35]	DBSCAN[k] clustering distributed topology analysis method	Mobile Smart Devices	Real taxi trajectory dataset of 11,947 taxis in Beijing, 2970 taxis in Suzhou, and 5450 taxis in Hangzhou, China	Best suitable solution in densely populated areas Increase in trajectory points yields an increase in efficiency	Distributed [23]DBSCAN didn't effectively address the growth in future data

Table 2 (continued)

Reference	Algorithms/methods	Sensors	Data	Results	Limitations
[36]	STS[l] method Non-linear programming model, mixed-integer linear programming model	Probe Vehicles	Data is collected from 5 trajectories for 50 vehicles in Jingshi Rd. and Shanghai East Rd. in Jinan City, Shangdong, China	Oversaturated phase number and delay time are minimized	Impacts of residual queues are ignored

This table covers algorithms, data collection places/sources, sensors used for data collection, and results in terms of accuracy and outcomes

[a]Stochastic partial differential equation method

[b]Automated Control Traffic Congestion System

[c]Cell Transmission Model

[d]Particle swarm optimization

[e]Sampling Rate matching

[f]Extended Cell Transmission Model

[g]Space-Temporal Congestion Subgraph

[h]Adaptive Smoothing Method

[i]Breadth-First Search

[j]Floating Car Data

[k]Density-based spatial clustering of applications with noise

[l]Sampled trajectory sample

analytical system for extracting traffic jam information through traffic speed for different road segments. Qi et al. [27] adopted an extended cell transmission method to reduce the incident-based traffic congestion.

Chen et al. [28] developed an interactive model for congestion propagation in between two segments by the Space-Temporal congestion subgraphs (STCS) method. Chen et al. [29] presented a rolling horizon scheme for heterogeneous traffic data to predict the traffic states for avoiding congestion by short-term traffic flow. Li et al. [30] developed an algorithm to recognize congested road segments and build congestion propagation graphs as shown in Figs. 7 and 8. An innovative approach was adopted by integrating PageRank and clustering algorithm to predict the effect of rainfall in urban traffic [31]. Balal et al. [32] used the dynamic user equilibrium method to develop resilience curves and correlation analysis to minimize the disruption on the urban highway caused by an incident. However to find the exact location of congestion a method was proposed based on real-time processing of trajectory data in congested areas [33]. Wanjing et al. [34] an optimized framework was developed for vehicle trajectories to avoid congestion at intersections.

3.3 Vehicular Sensing Network Approaches

Urban traffic congestion can be effectively control by appropriate monitoring systems. Different approaches have been adopted as summarized in Table 3 to address traffic congestion

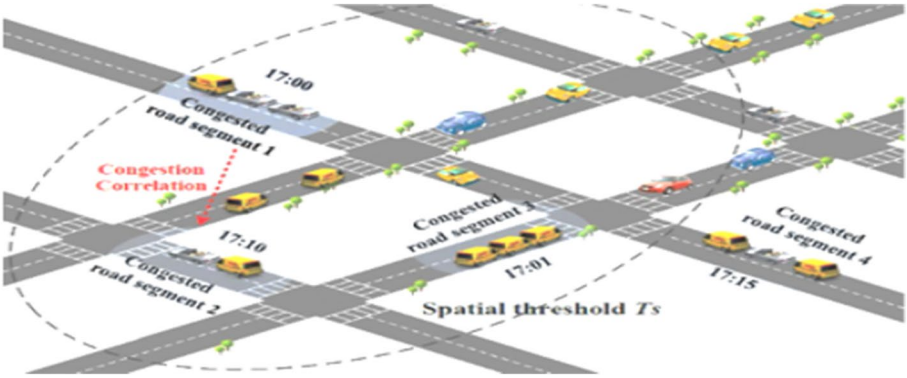

Fig. 7 Congested correlation between road segments at different observation time [30]

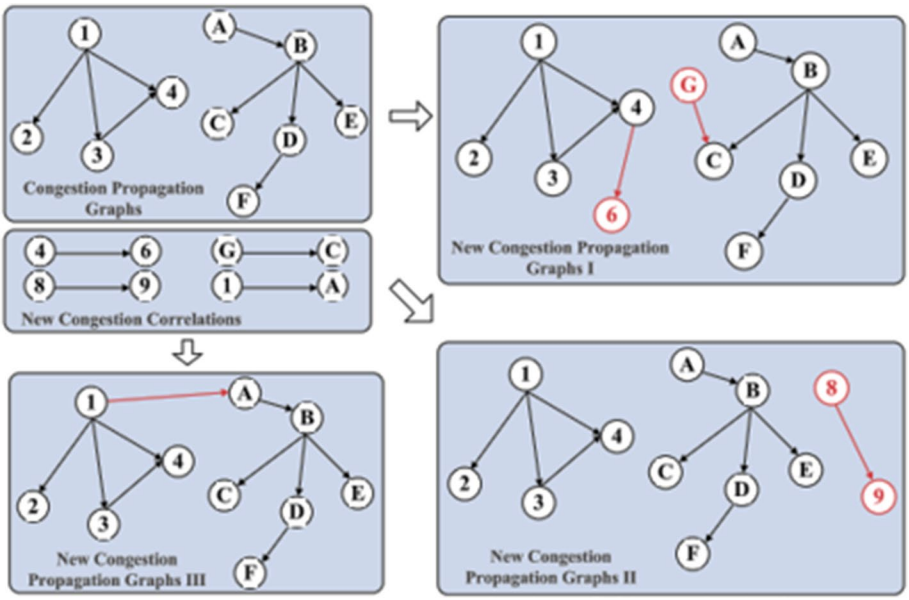

Fig. 8 Construction of congestion propogation graphs [30]

problems through vehicular sensor networks that can share and communicate information. The infrastructure of VSN is illustrated in Fig. 9 [35].

To share the effective information to the desired destination efficiently within a minimum time period an automated routing algorithm was proposed that decreased the delivery time to 30% thus improving the efficiency of communicating vehicles [36]. Dragoi et al. [37] developed a vehicular ad-hoc network for sharing congestion information in between connected vehicles that significantly reduced the travel time. Skordylis et al. [38] explored the solutions in-vehicle sensor networks for existing traffic infrastructure. To address the problem of spontaneous traffic congestion on the highway approach of the vehicular ad-hoc network was proposed that produces the major impact on traffic flow and reduced

Table 3 Summary of vehicular sensor network approaches for urban traffic road congestion for last ten years

Reference	Algorithms/methods	Sensors/communication equipment	Data	Results	Limitations
[38]	LA[a] Hybrid mobility model CLARA[c]	GPS[b], storage device digital map, wireless transmitter, and video camera	Data Collection place not mentioned	Network delay is reduced to 30% Packet delivery ratio is increased to 20%	No solution for real test beds Lack of cache maintenance and consistency
[39]	Traffic Model for congestion avoidance	Smartphones, laptops, or car computers	Data is collected from urban roads in Romania	Reduction in travel time up to 40% Reduction in emission up to 20%	No comparison of simulation based results with real life observations
[40]	D-Greedy and D-min cost[d]	GPS[b], Digital street map	Data is collected in dense urban areas of Zurich, Switzerland	The delivery delay is low Increase in packet delivery ratio Reduction in cost of communication in for monitoring	Real time traffic optimization is missing for nonrecurring traffic events
[41]	[29]VANET[e] strategies	Loop detectors	Empirical data collection of sudden breakdown from German Autobahn A044 between Unna and Werl	Comparatively, drop in case of communication from 22 to 12%	No provision for direct broadcasting of messages
[42]	Greedy patrol and circuit patrol algorithm	ON board sensors: accelerometer, GPS[b] receiver, and speedometer	Data is gathered from private vehicles and floating cars in Shanghai, China	Significant reduction in traffic estimation error from 35 to 10%	Low intensity of traffic, unknown traffic and varying traffic conditions are not sampled
[43]	SCADA[f] method DTCA[g]	ZigBee Based wireless network	Data Collection place not mentioned	Reduction in end-end delay Traffic flow is smooth	Large scale traffic flows are not considered
[44]	Neighboring Analysis	Radar, planar microscopic array antenna	Predefined Dataset used that is collected from Yangzijiang Road, Yangzhou City of China*** Experimental setup installed for 7.5 m road	Noise points are reduced to 4.1% from 239.9%	Radar data for different weather conditions are not tested

This table covers algorithms, data collection places/sources, sensors/communication equipment used for data collection, and results in terms of accuracy and outcomes

[a]Learning Automata-based routing scheme

[b]Global Positioning System

[c]Collaborative Learning Automata-based Routing Algorithm

[d]Data spreading algorithms

[e]Vehciluar Adhoc Network

[f]Supervisory Control and Data Acquisition

[g]Dynamic traffic cellular automata

**City/cities names not mentioned

***Dataset used: see [43]

Table 4 Summary of scheduling approaches for urban traffic road congestion for last ten years

Reference	Algorithms/methods	Data	Results	Limitations
[46]	EDF[a] FP[b]	Data Collection place not mentioned	EDF[a] algorithm is more efficient than the [33]FP algorithm Increasing number of lanes does not affect the performance of the scheduling algorithm	No provision for low priority vehicles
[47]	Flooding-based geo-cast method, T-window average, N-window average	The downtown area of Tianjin for road segment of 2.2 km in between Hebei and Haiguangsi intersection	Average travel time for buses are used to describe the congestion condition	Does not applicable where buses travel on reserved lanes Reliability and timeliness can be further improve
[48]	DCF[c] PCF[d]	Data Collection place not mentioned	Significant decrease in average time to pass an intersection	Mechanism are not adequate to support real time traffic
[49]	DCPM[e] SCPM[f]	Data extracted from the reported past study in Northern Taiwan	A stochastic model can solve large size traffic data more effectively	Multiple stochastic factors are not considered
[50]	MDDF[g] MADDF[h]	Data Collection place not mentioned	MDDF[g] and MADDF[h] shows promising results for complex intersections with high traffic density	MDDF and MADDF doesn't show promising results for simple intersections
[51]	KNR[i]	Data is collected from a linking road of national highway to lower level road network in the city of Hague, Netherland	KNR[i] successfully reduced delay time	Not generalized effect due to limitation of network size
[52]	WPM[j]	Congestion factor found to be 1.4 and value above it will belong to NRC	Congestion factor found to be 1.4 and value above it will belong to Non-recurrent congestion	Traffic flow on road networks are not considered
[53]	Nash equilibrium	Data from two road intersections for 50,100 and 150 cars have been collected**	Network performance is increased in terms of channel busy time to 42.27%, message loss rate to 43.94%, and data collision to 47.66%	Not appropriate checking for saturation condition of transmitting channel

Table 4 (continued)

This table covers algorithms, data collection places/sources, and results in terms of accuracy and comparisons with other models

[a]Earliest deadline first algorithm

[b]Fixed Priority algorithm

[c]Distributed Coordination Function

[d]Point Coordination Function

[e]Deterministic carpooling model

[f]Stochastic carpooling model

[g]Minimum destination distance first

[h]Minimum average destination distance first

[i]Kiss and Ride

[j]Weighted Product Model

[**]City name omitted by the author

travel times [39]. Sensing vehicles on roads have limited coverage; a patrol algorithm was proposed to estimate un-sampled traffic data and yields that the traffic estimation error is reduced to 10% for controllable floating cars [40]. Eren et al. [41] developed a method to avoid traffic congestion by determining the traffic throughput on urban roads. Liu et al. [42] proposed a solution for abnormal traffic data; a neighboring analysis method is adopted to extract the similar characteristics from neighbor roads, finely tuned them, and then predict traffic flow.

3.4 Scheduling Approaches

The problem of traffic congestion is greatly affected by the delay and lack of proper scheduling mechanisms. Diverse approaches have been adopted for effective and early prediction traffic congestion using scheduling approaches as indicated in Table 4.

Ahmad et al. [44] performed a comparative analysis for [32]EDF and [33]FP algorithms for a reduction on unwanted delay in traffic flow and found that [32]EDF analysis is more efficient than [33]FP method. Congestion caused by buses can affect the traveling times of vehicles on road; so a novel detection scheme was proposed by evaluating the travel time of buses and identify the congestion status [45]. Park et al. [46] utilized the concept of contention-free urban ways and find a balanced efficiency in between traffic flow and users. A simulation-based car polling method was constructed for stochastic travel times of vehicles on roads to address the different traffic flow problems [47]. Ahmad et al. [48] developed a solution to reduce congestion at intersections by conventional scheduling method and a significant reduction of around 80% in traffic congestion is observed. Overtoom et al. [49] proposed a simulation-based method to reveal the effect of congestion of shared autonomous vehicles and decrease in delays are observed. A method for prediction of non-recurrent congestion was proposed by observing the high link journey time estimations (LJTs) on neighboring roads and weighted analysis showed that LJTs greater than 40% are non-recurrent congestion [50]. Amer et al. [51] developed an approach for the Internet of vehicles to improve the efficiency of message transmission in case of a dense traffic environment.

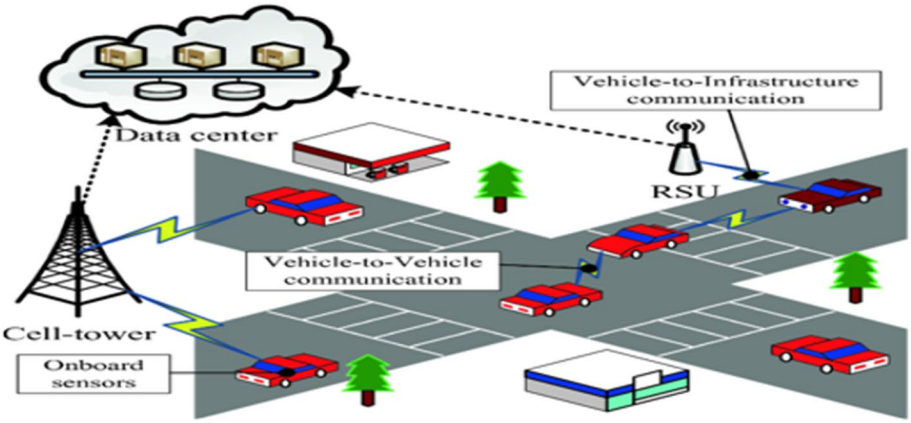

Fig. 9 Communication based VSN network for congestion prediction [35]

3.5 Artificial Intelligence Approaches

Prediction of traffic congestion has shown remarkable improvement by implementation of Artificial Intelligence (AI) approaches see Fig. 10 [52]. Many researchers are still working on different branches of AI to handle the problem of congestion as illustrated in Table 5.

Zhang et al. [53] proposed an intelligent prediction system that provides real-time prediction results for heterogeneous input data that occurs due to incidents on urban roads. An identification model was developed to predict the four levels of traffic states blockage, congestion, free slow, and slight congestion by proposing the fusion of two algorithms k-mean and neural network [54]. Li et al. [55] proposed a method that performs quantitative analysis in congested traffic conditions and performs well for different vehicle features and weather conditions. El-Tantawy [56] an adopted approach to handle traffic congestion is by adjusting timing plans for traffic signals at intersections. An innovative approach of artificial neural network was adopted to predict the traffic flow by combining geographical and environmental factors and performed adoptive forecasting of congestion on urban roads [57]. Kim et al. [58] adopted an approach to avoid the problem of dimensionality by estimating the stochastic travel times represented and through the rollout approach, the good solution is adopted instead of the optimal solution. Al Mallah et al. [59] proposed a scheme for classification for urban non-recurrent congestion by adopting four different techniques of machine learning and Naïve Bayesian found to be most accurate for prediction of non-recurrent congestion. A method histogram-based approach along with a rerouting strategy for prior prediction of congestion so that proactive actions can be taken to avoid congestion [60]. An innovative approach by using a deep convolution neural network was developed to capture the characteristics of multiscale problems in different levels of traffic congestion [61]. A visual analytical system was developed by volume-speed rivers (VSRivers) to explore the different causes of congestions and their severity by observing traffic speed and volumes [62]. Zheng et al. [63] proposed a deep learning approach that enhanced the decision-making ability in traffic management. Wang et al. [64] developed a cooperative control method with a reinforcement learning approach for optimization of mobility in freeways and shows the early response for any congestion condition.

Zheng et al. [65] proposed a deep and embedding learning method by integration of embedding component, LSTM, and CNN for congestion prediction based on fine-grained

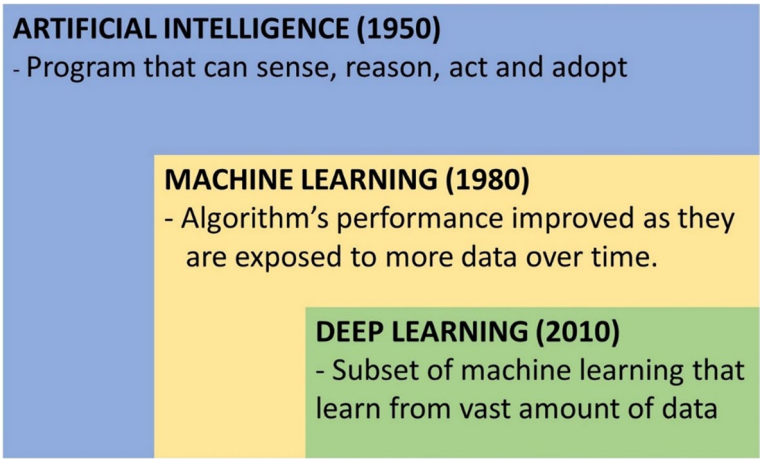

Fig. 10 Artificial Intelligence Prediction [52]

traffic states. A novel algorithm was developed for short-term traffic flow prediction that adopt the change in real-time traffic data and predicted the accurate result [66]. Zhao et al. [67] proposed a temporal convolution network to capture the temporal and spatial evolution of traffic flow with forecasting accuracy of 94.52%. Rajan et al. [68] proposed a hybrid neural network for prediction of congestion level with improved computational efficiency and in comparison PredNet have high precision ranges than ConvLSTM and Autoencoder techniques. Xu et al. [69] developed a system to analyze hotspots in a region that may be exposed to traffic congestion. Masud et al. [70] proposed two techniques for successful detection and counting of vehicles on sunny and cloudy days among which HOG based approach yields good results for the F-measure score of 0.9844 and 0.9949 for two different images sequences. Pi et al. [71] developed a method to identify the causes of congestion through N-curves by observing the traffic flow in congested areas.

Abdi et al. [72] proposed an algorithm that increases the forecasting accuracy and decreases the computational time by combining the linear and non-linear traffic models. The three-dimensional space model was developed to classify the different traffic flows through SVM with good accuracy [73]. However to predict and control the traffic flow in case of incident occurrence on-ramp an approach was proposed that increase the prediction accuracy to 92.5% [74]. An innovative approach of classical hesitant distances and hesitant distances sets (HDS) was proposed for urban traffic state identifications [75]. Zhang et al. [76] adopted an approach using Generative Adversarial Network for prediction of traffic flows for different road links, GAN combined with different ML techniques as illustrated in Fig. 11 to gain significant improvement in the prediction of traffic flow. Zhang et al. [77] developed a short-term traffic flow system with three-dimensional traffic data that have high prediction accuracy for diverse traffic conditions. A deep learning approach with non-parametric analysis was performed for the evaluation of traffic congestion in urban areas [78]. Tian et al. [79] method of traffic prediction was developed to overcome the limitation of static spatial dependency that improves the prediction accuracy. An efficient method for traffic congestion prediction is adopted on developed image dataset with high accuracy [80]. An intelligent fusion based machine learning approach is adopted for congestion prediction with high accuracy of 95% [81].

4 Discussions

Traffic congestion is globally accepted as one of the major hindrances in the infrastructure of any country. In between 2019 to 2020 due to COVID-19 Pandemic traffic congestion is reduced in 387 cities but increased in 13 cities as facts and figures illustrated [83]. In the year 2020 analysis has been performed for 415 cities and found that these congestion rates are increasing and decreasing in different cities according to their response in COVID-19 condition [83]. Many predictions and forecasting methods have been already adopted as described in Tables 1, 2, 3, 4, and 5.

The heuristic approaches have provided optimal solutions for significant problems in traffic management for any inappropriate network design and make decisions simpler and faster through good calculations [13–17, 20, 22]. But constrain is that they can lead to poor decision-making due to the availability of limited datasets [17, 19, 21].

Data sensing approaches have a lavage of the large coverage area for recording different traffic parameters efficiently and can be interpreted and analyze through different techniques that have good accuracy in real time prediction [31, 32, 35]. These approaches are expensive and time-consuming methods, especially when deployed in small areas [24, 28, 33].

Vehicular sensing techniques (VSN) have shifted the paradigm of gathering traffic parameters in environments where there are constraints of energy consumption. So VSN provides better solution because the vehicles don't have energy problems. The VSN gain its popularity where there is a massive gain in traffic data over time periods with fast transfer of data [38, 40–44]. This approaches can be affected by unreliable cellular communication links for transmitting data but disposed to network interfaces [41].

Scheduling algorithms are significant for priority-based predictions and reducing the delay while transmitting any relevant information to concerned traffic authorities [46, 49, 51]. They normally have low response time but can have poor average time while transmitting a great amount of traffic information [47, 48, 53].

Artificial intelligence approaches are valuable for the prediction of traffic congestion that can provide real-time congestion prediction, short term, and future traffic prediction [55, 56, 58, 59, 62–67]. They are adaptable for large amounts of data sets including traffic and environmental parameters in congested areas [55, 57, 59–61]. Yields good adaption for dynamic characteristics of traffic and predict high accuracy rates [61, 69, 72, 73, 78–81].

5 Conclusion

Accurate and successful prediction techniques for traffic congestion is key in the development stages of any country. A significant amount of research work has been done to address this issue. In this work, a persistent effort has been made to summarize different traffic congestion prediction techniques into five major categories. Congestion prediction techniques have been discussed for each category with their limitations and results. Among all these approaches, Machine learning algorithms yield the best result, especially with the implementation of advanced deep learning algorithms such as Generative Adversarial Networks (GAN) because of their remarkable ability to learn in different environments. Overall, this study identifies the challenges and GAP in congestion prediction techniques and provides new visions for researchers to develop more sustainable and accurate traffic prediction systems.

Table 5 Summary of artificial intelligence approaches for urban traffic road congestion for last ten years

Reference	Algorithms/methods	Data	Results	Limitations
[55]	FNN[a]	Data is collected from an incident in Guanghua road on 9th December 2008 at 18.05	Valuable for well cooperated for online and offline systems	Approximation missing to reduce the impacts of congestion
[56]	Fuzzy fusion algorithm	190 thousand data set is collected from Floating car data in Beijing's Expressway, China	Fusion algorithm yields accuracy of 91.6% over other models	Different parameters speed limitations, heavy vehicles, bottleneck and waiting time are not addressed
[57]	Agent-based modeling	Data Collection place not mentioned	Overcome the occlusion problem in congested areas	Time consuming approach Cannot apply on real time traffic monitoring
[58]	MARLIN-ATSC[c]	Data is collected and implemented at 59 intersections in the downtown city of Toronto, ON, Canada	Average intersection delay is reduced to 27% and travel time savings is 15% Average intersection delay is reduced to 39% on the network level and travel time saving is 26%	Model reveals higher delay for latter data experiences
[59]	GN-ANN[d] LM-NN[f]	Data and road information is collected in the city of Suwon, South Korea	MPRM[e] shows better performance over other methods with good results and with fewer prediction times	Accidental conditions and huge structured buildings are not considered for database
[60]	Markov decision process	Data is collected from the delivery company in Singapore with multiple customers	Roll out approach improves the travel time to 7%	Rollout approach ignored states with low error rate
[61]	Naive Bayesian algorithm	Synthetic data were collected from 700 000 individual car trips in the city of Cologne, Germany	Nave Bayesian have high accuracy prediction of 89.51%	Lack of more sophisticated algorithm must be cooperated for connected vehicles in real world operations
[62]	Histogram based approach	Data is collected in the city of Al Ain, UAE	Prediction of source volume is reduced to 10% of the time Reduction rate of 5.35% to achieve required link speed Reduction of 4.7% in link length	No generalized method of identification of a congested lane due to different characteristics of lanes
[63]	PCCN: Deep convolution neural network	Vehicle passage record of six weeks was collected from 614 road segments in the city of Jinan, China	Error rate for MRE[g] is reduced by 2.8% Error rate of MLP[h] is reduced to 5.4% PCCN performance outclasses the different state-of-art algorithms	No predictions for long term traffic congestion
[64]	LSTM[i] recurrent neural network	Data is collected from experts from the congestion management center for 18 months from taxi drivers and 128 CCTV cameras	LSTM[i] (S, T, R) yields good accuracy of 31.5% Quantitate and qualitative evaluation with good accuracy	Accident and weather conditions are not part of database

Table 5 (continued)

Reference	Algorithms/methods	Data	Results	Limitations
[65]	LSTM[i]	Pre-processed data from OpenITS that was collected from No. 1 Yuanda Road, Changsha, China	LSTM[i] shows more accurate results than ARIMA[j] and BPNN[k] model	Prediction accuracy is less for long time step
[66]	DQL[l]	2 km freeway was selected as Testbed in Drechttunnel, Netherland	Modified DRL[m] effectively reduce traffic congestion Speed differences are significantly reduced to lower the collision risks Inherent flaws are tackled by RL	No provision of control for ramps on freeway
[67]	LSTM[i] CNN[n]	Dataset provided by Knowledge Discovery and Data Mining Tools completion (KDD CUP 2017) in China**	Travel time prediction in 0.5 s Features of CNN[n] in the DELA model help to perform well on a large scale	Poor interpolation ability Learning ability is limited
[68]	Adaptive kernel Support vector machine	Freeway in District 10, San Joaquin County, California, USA	Better adoption of dynamic characteristics of urban traffic	Delay in simulation results in comparison with real values
[69]	DCNN[o]	Data is obtained from RSU for every 15 min in Birmingham, UK	Higher accuracy of 95.42% over LSTM[i] and CNN[n]-LSTM[i]	No fluctuations in heavy traffic is considered
[70]	CNN[n] LSTM[i] TCNN[p]	Data gathered from congestion map provided by online web service TOPIS for central Seoul, South Korea	Performance gain is about 12.5%	Proposed model is not generalized for large-scale with sustained trainability
[71]	PR K-Clustering[q]	Taxi trajectory data is collected in Shanghai, China	Faster training speed than LSTM Prediction time is 0.01 min	Overlapping of evaluated road segments with other roads are not identified that may affect the damping factor
[72]	HOG[r] Foreground Estimation	GRAM-Road Monitoring dataset	Accuracy is about 0.9844 for a sunny day Accuracy is about 0.9849 for a cloudy day	No real time prediction for changing weather conditions to avoid congestions states
[73]	CNN[n]	GPS dataset is collected from buses and taxis in the city of Beijing from China, Seoul from South Korea, and San Francisco from the USA	Classification accuracy is about 92% through visual patterns	Visual system analysis didn't predict the real time traffic conditions
[74]	Neuro-Fuzzy Model	Real-time data is collected from 2-lane eastbound freeway in Minnesota, USA	Best prediction for traffic flow and prevent the overfitting problem efficiently	More error indexes for ARIMANN-TS Neuro Fuzzy based algorithm
[75]	SVM[s]	Observational set of 200 dataset**	SVM[s] kernel produces good results Polynomial kernel function works well for un-normalized data	Unnormailized classification is not satisfactory for sigmoid kernel function

Table 5 (continued)

Reference	Algorithms/methods	Data	Results	Limitations
[76]	BNN[k]	Observational set of 600 data**	Reduced the prediction delay by 25% with a variance of 0.002 Accuracy is about 92.5%	Peak time parameters are ignored
[77]	FNN[a]	Data collected in China**	Proposed model is good in environments of objective application	Not suitable for subjective application environments
[78]	CNN[n] LSTM[i] GAN[t]	GPS data is collected from road networks in the city of Chicago and Bay of California, USA	TrafficGAN[t] outclass the other statistical models	Multi-step flow predictions for congestion scenario is not addressed
[79]	RGCNN[u]	Data is collected from the Caltrans performance measurement system (PeMS) that collect data from 12 consecutive locations	Prediction model has higher accuracy and less calculation time Predicted model can be adopted in different traffic condition	Network based prediction is not performed that decreases the reliability of information sharing
[82]	FNN[v]	GPS data is collected from 11,033 taxis Xian South Second Ring Road, China	Most suitable parameter is accelerated failure time Quality of data is improved and congestion prediction with high reliability	Environmental conditions are not considered that may directly affect the long term traffic planning routines

This table covers algorithms, data collection places/sources, and results in terms of accuracy and comparisons with other models where available

[a]Fuzzy neural network

[b]Support Vector Regression

[c]Multi-agent reinforcement learning-Adaptive traffic signal control

[d]Gaussian model with an artificial neural network

[e]Multifactor Pattern recognition model

[f]Levenberg-Marquardt with back-propagation

[g]Mean Relative Error

[h]Multilayer perceptron

[i]Long short-term memory

[j]AutoRegressive Integrated Moving Average

[k]Backpropagation neural network

[l]Distributed Q Learning

[m]Deep Reinforcement Learning

[n]Convolution neural network

[o]Deep convolution neural network

[p]Transpose Convolutional Neural Network

[q]PageRank K-clustering algorithm

[r]Histogram of Oriented Gradients

[s]Support Vector Machine

[t]Generative Adversarial Networks

[u]Double-branch deep residual gated convolutional neural network

[v]Feedforward neural network

**City name omitted by the author

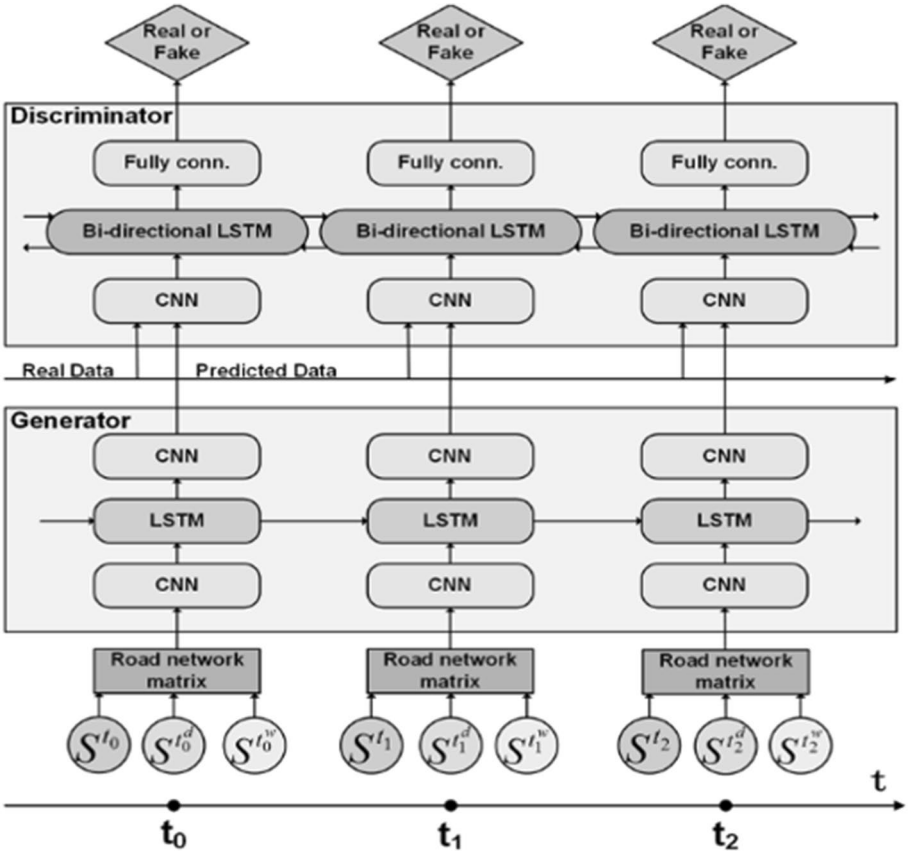

Fig. 11 TrafficGAN architecture shows the implementation of discriminator and generator [76]

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Declarations

Conflict of interest There is no conflict of interest.

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